WASP-1b

R-1146

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_191004` · created 2026-07-31T00:40:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458760.885 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.041
Transit depth (Rp/Rs)^2	1.91 +/- 1.13 [%]
Orbital Inclination (inc)	89.42 +/- 1.47
Airmass coefficient 1 (a1)	2.3 +/- 1.25
Airmass coefficient 2 (a2)	-0.34 +/- 0.57
Scatter in the residuals of the lightcurve fit is	57.28 %
Best Comparison Star	None
Optimal Aperture	0.9
Optimal Annulus	3.6
Transit Duration (day)	0.059 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.041 vs 0.1036 (deviation 0.54σ)
Duration fit vs published	0.059 d vs 0.1567 d (deviation 5.74σ)
Mid-time O–C	-10.3 min
Depth SNR	2.2
Residual scatter	57.28 %

Light curve

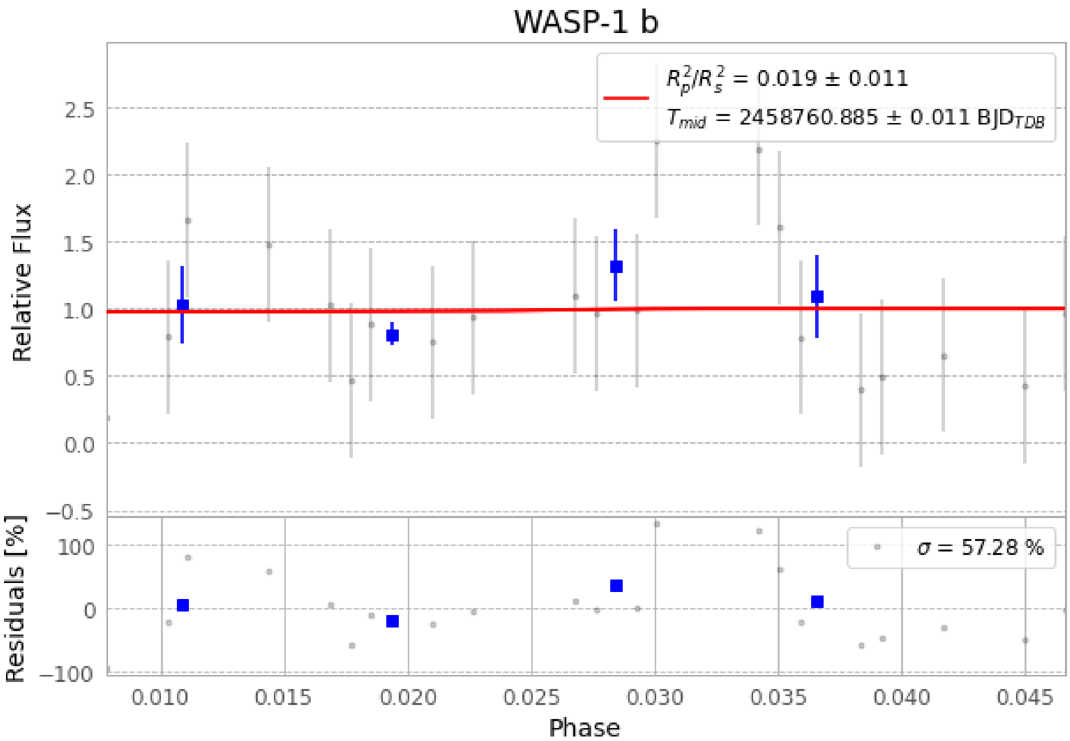

Figure 1 — FinalLightCurve_WASP-1 b_2019-10-04.png (SHA-256 3d8b8368f88976f8…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [231, 262], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 371c183842efe794…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

53 FITS frames (35 MB), `WASP-1191004092412.FITS` → `WASP-1191004120012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-191004.log (402562bbed1d…), FinalLightCurve_WASP-1 b_2019-10-04.png (3d8b8368f889…), FinalLightCurve_WASP-1 b_2019-10-04.csv (4d0e8d568767…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)